LAB programs cycle 2

1) DES

Data Encryption Standard:

Aim:

To implement Data encryption standard for encryption and decryption using secret key

Algorithm:

* Importance the DES cipher and padding function
* Read the plaintext 8 byte secret key
* Read plaintext to multiple 8 bytes
* Encryption plaintext using DES in ECB mode
* Decrypt ciphertext display original text

Code:

```
DES - Key Schedule ks;
DES -cblock input,output
DES - Set key unchecked (&key, &ks);
DES - ecb. encrypt (&input, &output, &ks, DES.
                    Encrypt)
```

Input

plaintext: Hello 123

Key: 12345678

output:

Encrypted: ciphertext in hexadecimal

Decrypted: Hello123

Result:

Thus the c program demonstrated successfully.

2. 2 DES

Aim:

To demonstrate double encryption using DES key

Algorithm:

* Read the plaintext
* Read two keys
* Apply first key
* Apply second key
* Display ciphertext

code:

```
#include <stdio.h>
int main() {
int P, K1, K2;
scanf("%d %d %d", &P, &K1, &K2)
Print("ciphertext = %d", P^K1^K2);
return 0; }
```

Input: 10 5 3

ciphertext = 12

Result:

Double encryption was demonstrated successfully.

Triple DES

Aim

To demonstrate Triple encryption using operations.

Algorithm:

* Read plaintext
* Read the key
* Apply encryption operation
* Repeat the operation
* Display ciphertext

Code:

```
#include <stdio.h>
int main() {
int p, k; scanf("%d%d", &p, &k)
printf("ciphertext = %d", (p^k)^k^k);
return 0; }
```

Input: 10 5

ciphertext = 15

Result:

Triple encryption was demonstrated successfully.

4. Blowfish Algorithm.

Aim:

To demonstrate to simple substitution operation

Algorithm.

* Read the plaintext
* Read the key
* perform arithmetic transformation
* Generate ciphertext
* Display the result.

Code:

```
#include <stdio.h>
int main() {
int p, k; scanf("%d%d", &p, &k)
Printf("ciphertext = %d": (p*5+k),
return 0; }
```

Input: 10 5

ciphertext - 55

Result:

Simple encryption transformation was demonstrated successfully.

5. Rc5 Algorithm:

Aim:

To demonstrate a simple Rc5 style transformation

Algorithm:

* Read plaintext
* Read the key
* perform addition
* Apply XOR operation
* Display ciphertext.

code:

```
#include <stdio.h>
int main() {
int p, k; scanf("%d%d", &p, &k)
```

```
Printf("ciphertext = %d", ((p+k)%256)^3);
return 0;}
```

Input = 10 5

ciphertext = 12

Result:

Simple encryption transformation was demonstrated successfully.

6. Modes of DES

Aim:

To demonstrate ECB and CBC encryption modes using XOR.

Algorithm:

* Read two plaintext blocks
* Read the key and IV
* perform ECB operations
* Perform CBC operations
* Display encrypted blocks

Code:

```
#include <stdio.h>
int main () {
int P1, P2, K, IV;
Scanf ("%d %d %d %d", &P1, &P2, &K, &IV);
Printf("ECB = %d %d In CBC = %d %d;
                    P1^K) P2^K;
(P1^IV)^K, (P2^IV)^K))^K);
```

Input: 10 20 5 3

ECB: 15 17

CBC = 8 25

Result:

ECB and CBC modes were demonstrated successfully

7. RSA Algorithm

Aim:

To write a c program to demonstrate RSA encryption.

Algorithm:

* Read message and public key
* Read modulus value
* calculate modular exponentiation
* Generic ciphertext
* Display the result.

code

```
#include <stdio.h>
int main () {
int m,e,n,c=1,i; scanf("%d%d%d",
                    &m, &e, &n);
for(i=0; i<e; i++) c = (c*m)/n;
return 0; }
```

Input : 4 3 33

ciphertext : 31

Result:

RSA encryption was demonstrated successfully.

8. Key exchange Diffie Hellman

Aim:

To determine secret key exchange using Diffie-Hellman.

Algorithm:

* Select public values
* Read private key
* Calculate public key
* Generate shared secret key
* Display shared key

code:

```
#include <stdio.h>
int main(){
int g=5, a=6, b=15, p=23, x=1, y=1, i;
for(i=0; i<a; i++) x = x*g%p;
for(i=0; i<b; i++) y = y*g%p;
printf("Shared key = %d", y);
return 0; }
```

Shared key = 2

Result:

Diffie Hellman key exchange was demonstrated successfully